

TEJAS NATEKAR

Mechanical and Aerospace Engineering Student

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[Portfolio -](#)

EXECUTIVE SUMMARY

Mechanical and Aerospace Engineering undergraduate at Mahindra University with interest in thermal-fluid systems, computational modelling, and mechanical design. With an industrial exposure to aerospace manufacturing during an internship at AZAD Engineering with shop-floor exposure to CNC machining, CMM inspection, tolerance verification and precision measurement. Experienced with working exposure in MATLAB, SolidWorks, and ANSYS through projects, and seeking opportunities in mechanical design, thermal-fluid systems and aerospace engineering.

EDUCATION

Bachelor of Technology in Mechanical Engineering

August 2023 – June 2027

Mahindra University | École Centrale School of Engineering

- **Relevant Coursework:** Thermodynamics, Fluid Mechanics, Computational Fluid Dynamics, Material Science, Finite Element Method, Design of Machine Elements, and Mechanics & Dynamics.
- **Areas of Interest:** Thermal-Fluid Systems, CFD, Finite Element Analysis, Numerical Modelling and Study of Materials.

SKILLS

Programming Languages: C/C++, Python, MATLAB, CNC Coding, LaTeX

Tools: SolidWorks, Inventor, AutoCAD, ANSYS, LabVIEW, Microsoft Office

EXPERIENCE

AZAD Engineering Ltd. Hyderabad, Telangana

July – August 2025

Internship

- Observed the step-by-step manufacturing of aerospace turbine blades, focusing heavily on how 5-axis CNC machines operate, tooling practices, Lean Manufacturing practices and CMM tools on the shop floor.
- Learned about strict quality control procedures such as dimensional inspection, tolerance verification, and material inspection.
- Developed understanding of how machining strategy, tool selection, fixture rigidity, inspection methods, and process control directly affect final component accuracy and production reliability.

PROJECTS

Tribological Study of Bismuth Oxide Nanoparticles as Lubricant Additives

January 2026 – Present

Literature Review, Material Analysis

- Studying Bi₂O₃ nanoparticles as lubricant additives for reducing friction and wear in mechanical contacts.
- Studying how particle size, morphology, synthesis route, and material behaviour influence tribological performance.
- Learning how material selection at the microscopic level can affect friction, wear, lubrication stability, and component life.

Design and Analysis of a Tesla Turbine

January – April 2026

Academic Project – SolidWorks, CFD

- Designed and simulated a 12-disk bladeless turbine in SolidWorks; linked design choices to turbomachinery calculations.
- Identified what went wrong: CFD/assembly review showed casing bypass, choking/leakage risk and sealing sensitivity.

- Learnt that in real mechanical assembly, manufacturability, sealing, and tolerances are as important as the core working principle of the system i.e. fluid momentum transfer.

Machine Learning Model for Aerodynamics

January – April 2026

Academic Project – Python, Machine Learning

- Built a Random Forest model to predict non-linear lift and drag behaviour and classify stall boundaries from aerodynamic datasets, bypassing expensive CFD simulations.
- Used data preprocessing, thresholding, model training, and error analysis to reduce dependence on repeated CFD simulations for every design point.
- Learnt how machine learning can act as a fast surrogate model, but also that prediction quality depends strongly on data coverage and physical consistency.

Electro-Thermal (Joule Heating) Multiphysics Analysis of a Steel Plate

January – April 2026

Academic Project – ANSYS

- Simulated coupled electric-current and thermal behaviour in a steel plate with a central hole to study Joule heating and voltage-induced temperature rise.
- Analysed how voltage, resistance, time duration, heat generation, and boundary conditions affect current density and temperature distribution.
- Identified unrealistic thermal spikes due to unconstrained loading and heat-loss assumptions, strengthening understanding of physically valid Multiphysics simulation setup.

The Road to Electrifying India's Public Transport

February 2024 – February 2025

Literature Review, Energy Systems, EV Systems, EV-Mobility

- Conducted a review of technological, economic, policy, and infrastructure barriers affecting public-transport electrification in India.
- Connected EV adoption challenges with battery systems, charging infrastructure, cost feasibility, and public-transport dependency.
- Learnt how engineering solutions must also be evaluated through economic, policy, and implementation constraints before large-scale adoption.
- Delivered insights at a National Level Conference, with the formal manuscript currently under peer review.

Autonomous Underwater Vehicle

February 2024 – February 2025

University AUV Team

- CAD modelling and mechanical design of AUV structural components, focusing on compact subsystem placement and manufacturable layouts.
- Faced practical design constraints related to limited internal space, component mounting, assembly access, and integration with other subsystems.
- Learnt how mechanical design in a team project requires balancing strength, packaging, and manufacturing ease.

Study of Heat Transfer of a Roof

August – November 2025

Academic Project – MATLAB, ANSYS

- Modelled heat transfer through roof materials by considering conduction through the roof, convection with surrounding air and the air discharged by the AC vent, and radiation from the exposed surface.
- Compared temperature distributions for different material and surface conditions using MATLAB to understand how roof design affects indoor thermal performance.
- Applied heat-transfer principles to evaluate how boundary conditions, surface properties, and material selection influence temperature distribution and thermal performance in roof systems.

Numerical and Analytical Study of an Unsteady 1-Dimensional Heat Conduction in a Copper Rod

February – May 2025

Academic Project – MATLAB

- Developed a finite-difference MATLAB model to study transient 1D heat conduction in a copper rod and compared the numerical results with analytical trends.
- Observed that boundary-condition assumptions, time-step size, and spatial discretisation strongly affect accuracy and stability of the solution.
- Applied finite-difference methods to simulate transient heat conduction, validating numerical trends against analytical behaviour to evaluate accuracy, stability, and boundary-condition sensitivity.